STUDENT PLACEMENT PREDICTION AND SKILL GAP ANALYSIS

Machine Learning Project Report

Project Type	Classification and rule-based skill-gap analysis
Dataset	Campus Recruitment dataset
Model	Random Forest Classifier
Environment	Kaggle Notebook / Python / scikit-learn

Prepared as a summary of the completed machine-learning prototype, its evaluation, and its initial recommendation logic.

1. Introduction

Student placement is an important outcome for educational institutions because it reflects how well students are prepared to enter the workforce. Academic performance, aptitude, practical project experience, internships, communication-related skills, and participation in placement training may all provide useful signals when examining placement outcomes.

This project develops a machine-learning prototype that predicts whether a student is likely to be placed based on a set of academic and non-academic attributes. It also includes a rule-based skill-gap analysis component that highlights selected areas for improvement and provides simple recommendations.

The system is intended as a decision-support prototype for exploration and student development. Its predictions are not guarantees of employment and should not be used as the sole basis for high-impact decisions.

2. Problem Statement

Students may find it difficult to identify which areas of their academic profile or preparation need additional attention before placement season. A structured analysis can help summarize placement-related patterns in historical data and provide understandable starting points for improvement.

Objective: Build and evaluate a binary classification model for predicting PlacementStatus (Placed or NotPlaced), then combine its prediction with a rule-based skill-gap analysis to offer targeted development suggestions.

Project objectives:

- Inspect and clean the dataset, including removal of duplicate records.
- Preprocess numerical and categorical features in a reproducible pipeline.
- Train a Random Forest classifier and evaluate it on a held-out test split.
- Use stratified cross-validation to assess stability.
- Rank model features by importance and implement transparent, threshold-based skill-gap recommendations.

3. Literature Survey

Maurya, Hussain, and Singh (2021) studied student placement prediction using machine-learning classification algorithms and academic-performance attributes. Their work compared several classifiers, including Support Vector Machines, Gaussian Naive Bayes, K-Nearest Neighbours, Random Forest, Decision Tree, Stochastic Gradient Descent, Logistic Regression, and Neural Networks. The study discussed evaluation using measures such as accuracy, precision, recall, F1-score, confusion matrix, and ROC/AUC.

The study provides a relevant foundation for treating placement prediction as a supervised classification problem and for evaluating models with more than accuracy alone. It also highlights the usefulness of academic indicators in placement analysis.

The present project follows a similar overall approach but uses a larger campus-recruitment dataset with academic, aptitude, soft-skill, project, internship, certification, and placement-training attributes. It uses a Random Forest pipeline and adds a separate rule-based skill-gap component. Results should not be compared directly across studies unless their datasets, target definitions, and evaluation protocols are aligned.

4. Methodology

4.1 Data Inspection and Cleaning

The initial dataset contained 7,225 rows and 11 columns. The inspection found 43 duplicate rows and no missing values. After removing duplicates and resetting the index, the final dataset contained 7,182 rows.

4.2 Feature and Target Selection

The target variable is PlacementStatus, with two classes: Placed and NotPlaced. Ten input features were used: CGPA, Internships, Projects, Workshops/Certifications, AptitudeTestScore, SoftSkillsRating, ExtracurricularActivities, PlacementTraining, SSC_Marks, and HSC_Marks.

4.3 Preprocessing

The data was split using a stratified 80:20 train/test split. The training set contained 5,745 records and the test set contained 1,437 records. A ColumnTransformer and OneHotEncoder(handle_unknown='ignore') encoded the categorical fields ExtracurricularActivities and PlacementTraining; numerical features were passed through. Preprocessing and model training were combined in a scikit-learn Pipeline to keep transformations consistent.

4.4 Model Training

A RandomForestClassifier was trained with 200 trees, random_state=42, class_weight='balanced', and n_jobs=-1. The balanced class weighting was included to account for the difference in class frequencies.

4.5 Skill-Gap Analysis

The skill-gap component is rule-based rather than a separately trained machine-learning model. It checks a student's profile against heuristic thresholds: CGPA below 7.5, aptitude score below 80, soft-skills rating below 4.3, projects below 2, internships below 1, workshops/certifications below 1, SSC marks below 65, and HSC marks below 70. A flag is generated when a threshold is not met, and recommendations are based on the flagged areas. These thresholds are preliminary and have not been validated against employer standards.

4.6 Student Assessment

The assessment function combines the classifier's predicted class and predicted probability with the rule-based recommendations. Probability values are model outputs and have not been calibrated; they should be interpreted cautiously.

5. Dataset

The project uses the Campus Recruitment CSV dataset hosted on Hugging Face. The dataset includes academic marks and measures of student preparation and activity. The target class distribution after duplicate removal was 4,192 NotPlaced records and 2,990 Placed records.

Dataset source: Krooz/Campus_Recruitment_CSV on Hugging Face

Selected descriptive statistics are shown below.

Feature	Mean	Std. dev.	Min	Max
CGPA	7.70	0.64	6.5	9.1
Internships	1.04	0.66	0	2
Projects	2.02	0.87	0	3
Workshops/Certifications	1.01	0.90	0	3
AptitudeTestScore	79.41	8.14	60	90
SoftSkillsRating	4.32	0.41	3.0	4.8
SSC_Marks	69.13	10.44	55	90
HSC_Marks	74.40	8.89	57	88

6. Results

6.1 Held-Out Test Performance

The model achieved 78.91% accuracy on the held-out test set of 1,437 records. The classification report is summarized below.

Class / Average	Precision	Recall	F1-score	Support
NotPlaced	0.80	0.85	0.83	839
Placed	0.77	0.70	0.73	598
Macro average	0.79	0.78	0.78	1,437
Weighted average	0.79	0.79	0.79	1,437

The model identifies the NotPlaced class with higher recall (0.85) than the Placed class (0.70). This means it misses a larger proportion of actual placed cases than actual not-placed cases. The result supports further refinement if the priority is to identify students who will ultimately be placed.

6.2 Cross-Validation

Five-fold stratified cross-validation on the training data produced a mean accuracy of 79.11%, with a standard deviation of 1.13 percentage points. Mean recall for the Placed class was 71.40%, and mean F1-score for that class was 73.99%. The similarity between cross-validation accuracy and held-out test accuracy suggests reasonably consistent performance on this dataset, although it does not establish generalization to other institutions or future cohorts.

6.3 Feature Importance

The five highest reported feature importances were:

Feature	Importance
HSC_Marks	18.59%
AptitudeTestScore	17.14%
SSC_Marks	13.32%
CGPA	11.78%
SoftSkillsRating	9.34%

Feature importance indicates how much a feature contributes to the fitted Random Forest's decisions under its importance measure. It does not establish that a feature causes placement outcomes, and correlated features can affect the interpretation.

6.4 Example Assessments

- Example profile 1: predicted NotPlaced, with a model-estimated Placed probability of 31%; the rule-based analysis flagged workshops/certifications as an area to improve.
- Example profile 2: predicted Placed, with a model-estimated Placed probability of 85.5%; no specific gaps were flagged by the current rules.

These examples illustrate the system's output format only. They are not evidence of real-world placement outcomes.

7. Conclusion

The project demonstrates an end-to-end machine-learning workflow for student placement prediction using a Random Forest classifier. The data was cleaned, categorical variables were encoded in a preprocessing pipeline, and model performance was assessed using a held-out test set and five-fold stratified cross-validation. The model achieved 78.91% test accuracy and 79.11% mean cross-validation accuracy.

The project also integrates a transparent, rule-based skill-gap analysis that turns selected profile thresholds into development suggestions. This component makes the prototype more actionable, but its thresholds remain heuristic and require review with educators, career counsellors, and placement professionals.

Limitations and future work: Evaluate the model on independent data from other cohorts or institutions; compare alternative classifiers; inspect possible bias and data leakage; tune thresholds using evidence and expert review; calibrate predicted probabilities if they will be presented as probabilities; and monitor changes in placement practices over time.

Overall, the prototype meets its core technical objective as an exploratory decision-support tool. It should support, not replace, human judgement and student counselling.

8. References

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